velocities in different directions. Speed will not give the direction of motion.




Speedometer

## Momentum

The linear momentum or simply momentum of a particle is defined as product of mass with velocity. It is denoted as '$\vec{p}$'. Momentum is also a vector quantity.

$$\vec{p} = m\vec{v}.$$

The direction of momentum is also in the direction of velocity, and the magnitude of momentum is equal to product of mass and speed of the particle.

$$p = mv$$

In component form the momentum can be written as

$$p_x\hat{i} + p_y\hat{j} + p_z\hat{k} = mv_x\hat{i} + mv_y\hat{j} + mv_z\hat{k}$$

Here $p_x = x$ component of momentum and is equal to $mv_x$

$p_y = y$ component of momentum and is equal to $mv_y$

$p_z = z$ component of momentum and is equal to $mv_z$

The momentum of the particle plays a very important role in Newton's laws. The physical significance of momentum can be well understood by the following example.

Consider a butterfly and a stone, both moving towards you with the same velocity 5 m $s^{-1}$. If both hit your body, the effects will not be the same. The effects not only depend upon the velocity, but also on the mass. The stone has greater mass compared to the butterfly. The momentum of the stone is thus greater than the momentum of the butterfly. It is the momentum which plays a major role in explaining the 'state' of motion of the object.

The unit of the momentum is kg m $s^{-1}$

### EXAMPLE 2.24

Consider two masses of 10 g and 1 kg moving with the same speed 10 m $s^{-1}$. Calculate the magnitude of the momentum.

#### *Solution*

We use $p = mv$

For the mass of 10 g, m = 0.01 kg

$$p = 0.01 \times 10 = 0.1 \text{ kg m s}^{-1}$$

For the mass of 1 kg

$$p = 1 \times 10 = 10 \text{ kg m s}^{-1}$$

Thus even though both the masses have the same speed, the momentum of the heavier mass is 100 times greater than that of the lighter mass.

## 2.10 MOTION ALONG ONE DIMENSION

### 2.10.1 Average velocity

If a particle moves in one dimension, say for example along the x direction, then

$$\text{The average velocity} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}.$$



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